

PROBLEM STATEMENT #15 | Healthcare & Telemedicine

Pharmacy Expiry & Re-order Dispatch Engine

1. Problem Context & Overview

Hospital pharmacies need an automated stock management engine that tracks batch expiry dates, generates First-Expired-First-Out (FEFO) dispensing lists, and triggers automated purchase orders when stock hits threshold.

Target Stakeholders / Actors: *Pharmacy Clerk, Inventory Supplier*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.

Sample Acceptance Criteria: Pass: Oldest valid batch selected first; Fail: Newer batch assigned while older unexpired batch exists.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below safety threshold.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.